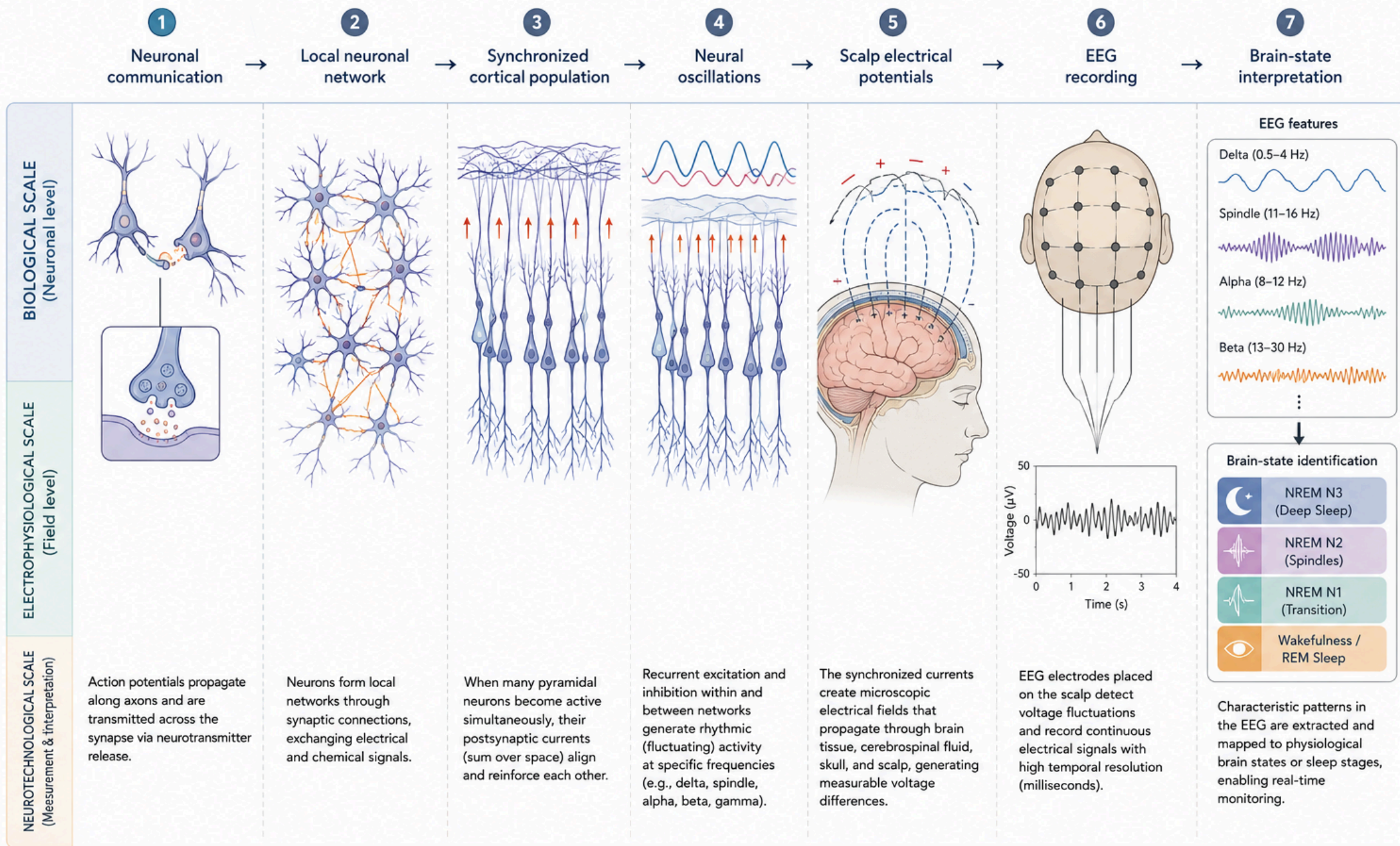




**Figure 3. From neurons to EEG and brain-state interpretation.** Neuronal communication at the cellular level scales up to synchronized activity in large cortical populations. The resulting oscillatory electrical fields propagate to the scalp, where they are detected by EEG electrodes. The recorded signals contain time-resolved information that can be used to infer the current brain state, including different sleep stages.